

MD ABDUR RABBI

Mechatronics Engineer | Robotics · Automation · Embedded Systems

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate of KUET (CGPA 3.57/4.00), seeking roles in automation, robotics, R&D/design, and embedded systems. Builds embedded and IoT systems on Arduino, ESP32, and STM32 with Python and Embedded C, backed by a final-year thesis comparing reinforcement-learning methods for safe robotic grasping in simulation. Aims to design and build automation systems for real industrial and manufacturing problems.

EDUCATION

Khulna University of Engineering & Technology (KUET) Mar 2022 – Aug 2026
B.Sc. in Mechatronics Engineering — **CGPA: 3.57 / 4.00**
Alamdanga Govt. College, Chuadanga — HSC (Science) — GPA: 5.00/5.00 2020
Alamdanga Academy, Chuadanga — SSC (Science) — GPA: 5.00/5.00, Govt. General Scholarship 2018

TECHNICAL SKILLS

Robotics & Simulation: NVIDIA Isaac Sim, Isaac Lab, computer vision, image processing (OpenCV)
Embedded & IoT: Arduino, ESP32, STM32 Nucleo, firmware, sensors & actuators, IoT dashboards
CAD & Simulation: SolidWorks, MATLAB, Simulink, Proteus, control systems
Programming: Python (NumPy, Matplotlib, OpenCV), C (Embedded C), Linux (basics)
Other: PLC (laboratory training), Microsoft Office (Word, Excel, PowerPoint)

PROJECTS

Undergraduate Thesis — Comparative Evaluation of Constrained and Unconstrained Deep Reinforcement Learning for Safe Robotic Grasping on a UR5e Manipulator in Simulation 2025 – 2026
• Simulation in NVIDIA Isaac Sim/Isaac Lab with RL policies on Linux.
Automated Paper Grading and Dispensing System — MTE 4108 Sessional, 5-member team project 2025
• Designed the NEMA17 lead-screw paper-feed mechanism (0.12 mm per sheet, sized for 100 GSM paper) and integrated Raspberry Pi control for OCR-based grading (Google Cloud Vision).
Gesture-Controlled Car — computer-vision-based control
• Built a car driven entirely by hand gestures: Python/OpenCV camera-based recognition translated hand movements into wireless drive commands for the microcontroller-controlled motor system.
IoT Real-Time Weather Monitoring Station — three independent platform builds
• Built a real-time weather station (temperature, humidity, pressure, rainfall, light) with DHT11, BMP180, rain, and LDR sensors; implemented separately on Arduino Uno, ESP32, and STM32 with a local LCD and a Wi-Fi IoT web dashboard.

EXPERIENCE

Core Organizer & Sector Lead — CALIBRATION 2.0, National Robotics Festival, KUET 2026
• Led Logistics & Kit Management segments for more than 1200 participants and organiser. Also led the RC Speed Battle segment; contributed to planning and budgeting and directed teams of junior volunteers.
Senior Executive, Drivetrain — Team Kilo Flight (Formula Student Team), KUET Jun 2024 – Sep 2025
• Worked on drivetrain design and electrical powertrain for one of Bangladesh's leading Formula Student teams.
Hardware Section Lead — KAVEN (Aircraft Design Team), KUET Nov 2024 – Feb 2025
• Led the hardware section for aircraft builds targeting international design competitions.
Member, Sports Committee — Residential Hall Committee, KUET 2024 – 2025
• Organized 5 tournaments for ~700 hall residents; maintained sporting facilities; served three terms as Hall Dining Manager.

AWARDS & ACTIVITIES

• **Champion** — Inter-Batch Badminton (Men's Singles), Dept. of Mechatronics Engineering; Inter-Department Tournaments 2026: key player (Badminton), team coach (Football), participant (Cricket).
Languages: Bangla (Native) · English (Fluent — speaking & writing)